THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** increases 100 times	Each unit of pH is a factor of 10.
2	**(a)** anhydrous sodium carbonate	Loses water of crystallisation.
3	**(d)** neutral	No colour change shows neutral.
4	**(a)** pH 1	Lower pH means more H^+.
5	**(d)** thermal decomposition	Heat decomposes baking soda.
6	**(c)** hydrogen	$2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$.
7	**(a)** HCl and NaOH	Strong acid + strong base gives neutral salt.
8	**(a)** disappears	Soluble calcium hydrogencarbonate forms.
9	**(c)** NaOH	NaOH ionises completely in water.
10	**(b)** basic	Strong base and weak acid give basic salt.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Strong acid ionises completely, e.g. HCl (1); Weak acid ionises partially, e.g. CH_3COOH (1).	1+1
12	Adds quick lime or slaked lime (1); Basic substances neutralise the acid (1).	1+1
13	Baking soda and tartaric acid release CO_2 (1); CO_2 bubbles make the cake rise (1).	1+1
14	pH 2 solution has more H^+ (1); 100 times as much (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	HCl: vigorous bubbling (1); CH_3COOH: slow bubbles (1); Water: no reaction; HCl is a stronger acid (1).	1+1+1
16	Electrolysis of brine (1); Anode: Cl_2; cathode: H_2 (1); NaOH remains in solution (1).	1+1+1
17	Acetic acid reacts with carbonate (1); $CaCO_3 + 2\,CH_3COOH \longrightarrow (CH_3COO)_2Ca + H_2O + CO_2$ (1); Bubbles of CO_2 (1).	1+1+1
	OR: Mix equal amounts of the same concentration (1); H^+ and OH^- react to form water (1); Neutralisation gives pH 7 (1).	1+1+1
18	It forms gypsum (1); Gypsum is a hard interlocking solid (1); Setting takes place in a short time (1).	1+1+1
	OR: Na_2CO_3 gives brisk effervescence (1); NaOH gives none (1); CO_2 turns lime water milky (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Litmus: HCl turns blue to red (1); NaOH and Na_2CO_3 turn red litmus blue (1); Add acid: Na_2CO_3 effervesces, NaOH does not (1); NaCl has no effect on litmus (1); Lime water confirms CO_2 (1).	1+1+1+1+1
	OR: $CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2}\,H_2O + 1.5\,H_2O$ (1); Heated at 373 K (1); $CaSO_4 \cdot \frac{1}{2}\,H_2O + 1.5\,H_2O \longrightarrow CaSO_4 \cdot 2\,H_2O$ (1); Uses: plaster, toys, decoration (1); Stored in moisture-proof container (1).	1+1+1+1+1
20	$Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); $Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (1); $NaOH + HCl \longrightarrow NaCl + H_2O$ (1); Bases with Zn: $2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$ (1); Acids and bases both conduct in solution (1).	1+1+1+1+1
	OR: Heat blue crystals in a dry test tube (1); Water drops condense on cooler walls (1); Residue is white anhydrous salt (1); Add water: blue colour returns (1); $CuSO_4 \cdot 5\,H_2O \rightleftharpoons CuSO_4 + 5\,H_2O$ (1).	1+1+1+1+1

Mistakes to avoid:

- Pouring water into concentrated acid; always add the acid slowly to water with stirring because the process is highly exothermic.
- Writing that a solution with pH 3 is 'less acidic' than one with pH 5; a lower pH means a **stronger** acid.
- Mixing up the formulae: baking soda is $NaHCO_3$, washing soda is $Na_2CO_3 \cdot 10\,H_2O$, plaster of Paris is $CaSO_4 \cdot \frac{1}{2}\,H_2O$ and bleaching powder is $CaOCl_2$.
- Thinking that all compounds containing hydrogen are acids; glucose and alcohol do not release H^+ ions.

- Writing H_2 as the gas formed when an acid reacts with a carbonate; the gas is CO_2.
- Assuming neutralisation always gives a neutral solution; a strong acid with a weak base gives an acidic salt solution (and vice-versa).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** formation of gypsum	Water is taken up.
2	**(d)** sulphuric acid	Lead-acid batteries.
3	**(c)** acidic	Strong acid and weak base give acidic salt.
4	**(b)** bleaching powder	$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$.
5	**(d)** high degree of ionisation	They ionise completely.
6	**(a)** basic	Turmeric turns red in bases.
7	**(a)** cathode	Positive H^+ ions go to the cathode.
8	**(d)** acetic acid	Acetic acid ionises partially.
9	**(a)** vinegar	Acid neutralises the alkaline sting.
10	**(b)** $NaHCO_3$	Hydrogencarbonates release CO_2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is a salt of a strong base and weak acid (1); Hydrolysis gives OH^-, making it basic (1).	1+1
12	Above 373 K it loses all water (1); Forms anhydrous $CaSO_4$, which does not set (1).	1+1
13	It is a mild base that neutralises stomach acid (1); $NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$ (1).	1+1
14	$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ (milky) (1); Milkiness disappears in excess CO_2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Base: any substance that neutralises acid, e.g. $Cu(OH)_2$ (1); Alkali: base soluble in water, e.g. NaOH (1); All alkalis are bases but not all bases are alkalis (1).	1+1+1
16	Releases CO_2 on heating (1); Makes dough rise (1); Neutralises acids in the stomach (1).	1+1+1
17	Plants grow best in a specific soil pH (1); Our body works in pH range 7.0-7.8 (1); Aquatic life depends on water pH (1).	1+1+1
	OR: Metal oxides are basic: $CuO + 2\,HCl \longrightarrow CuCl_2 + H_2O$ (1); Non-metal oxides are acidic: $CO_2 + 2\,NaOH \longrightarrow Na_2CO_3 + H_2O$ (1); Both give salt and water (1).	1+1+1
18	H^+ concentration decreases (1); pH increases (1); But never exceeds 7 (1).	1+1+1
	OR: Non-metal oxides are acidic (1); $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ (1); Salt and water form (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Stomach pH about 1.4 (1); Mouth pH below 5.5 corrodes enamel (1); Soil pH controls crop growth (1); Acid rain pH below 5.6 (1); Remedy using bases/antacids (1).	1+1+1+1+1
	OR: Acid: HCl (1); Base: $Cu(OH)_2$ (1); Alkali: NaOH (1); Indicator: litmus (1); Salt: NaCl (1).	1+1+1+1+1
20	Brine is electrolysed (1); $2\,NaCl + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$ (1); Cl_2: bleaching powder, PVC (1); H_2: fuel (1); NaOH: soaps, detergents (1).	1+1+1+1+1
	OR: $Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (1); $NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$ (1); $CuO + 2\,HCl \longrightarrow CuCl_2 + H_2O$ (1); $CO_2 + Ca(OH)_2 \longrightarrow CaCO_3 + H_2O$ (1); All form salt and water (1).	1+1+1+1+1

Mistakes to avoid:

- Pouring water into concentrated acid; always add the acid slowly to water with stirring because the process is highly exothermic.
- Writing that a solution with pH 3 is 'less acidic' than one with pH 5; a lower pH means a **stronger** acid.
- Mixing up the formulae: baking soda is $NaHCO_3$, washing soda is $Na_2CO_3 \cdot 10\,H_2O$, plaster of Paris is $CaSO_4 \cdot \frac{1}{2}\,H_2O$ and bleaching powder is $CaOCl_2$.
- Thinking that all compounds containing hydrogen are acids; glucose and alcohol do not release H^+ ions.
- Writing H_2 as the gas formed when an acid reacts with a carbonate; the gas is CO_2.

- Assuming neutralisation always gives a neutral solution; a strong acid with a weak base gives an acidic salt solution (and vice-versa).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** salt and water	$CuO + 2\,HCl \longrightarrow CuCl_2 + H_2O$.
2	**(c)** 7	Equal moles neutralise completely.
3	**(b)** 1.5	Two minus one half.
4	**(b)** aqueous sodium chloride	Concentrated NaCl solution.
5	**(a)** quick lime	Basic lime neutralises acidic soil.
6	**(a)** crystallisation	Water of crystallisation is taken up.
7	**(d)** carbon dioxide	CO_2 smothers fire.
8	**(a)** 10000	10^4 times.
9	**(a)** water	It turns blue with water.
10	**(b)** water of crystallisation is lost	Anhydrous salt is white.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It neutralises the extra acid (1); $Mg(OH)_2 + 2\,HCl \longrightarrow MgCl_2 + 2\,H_2O$ (1).	1+1
12	It forms hydrated $CuSO_4 \cdot 5\,H_2O$ (1); Hydrated crystals are blue (1).	1+1
13	$CaSO_4 \cdot \frac{1}{2}\,H_2O + 1.5\,H_2O \longrightarrow CaSO_4 \cdot 2\,H_2O$ (1); Gypsum (1).	1+1
14	HNO_3 is a strong oxidising agent (1); It oxidises the hydrogen to water and gives nitrogen oxides (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Acids give H^+ ions and bases give OH^- ions (1); Ions are mobile (1); Moving ions carry current (1).	1+1+1
16	$Na_2CO_3 \cdot 10\,H_2O$ vs $NaHCO_3$ (1); Washing soda loses water; baking soda gives CO_2 (1); Washing soda: cleaning; baking soda: baking (1).	1+1+1
17	Acids in them react with brass (Cu, Zn) (1); Poisonous salts form (1); Unsafe to consume (1).	1+1+1
	OR: Heating removes water of crystallisation (1); Anhydrous salt is white (1); Water gives back blue hydrated crystals (1).	1+1+1
18	$CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2}\,H_2O + \frac{3}{2}\,H_2O$ (1); Water lost = 2 - 1/2 = 1.5 molecules (1); Heated at 373 K (1).	1+1+1
	OR: $2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$ (2); $Na_2CO_3 + 10\,H_2O \longrightarrow Na_2CO_3 \cdot 10\,H_2O$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Prepared from brine, ammonia and CO_2 (1); $NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$ (1); Mild non-corrosive base (1); $2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$ (1); Uses: cakes, antacid, fire extinguisher (1).	1+1+1+1+1
	OR: NaCl: strong acid + strong base: neutral (1); NH_4Cl: strong acid + weak base: acidic (1); Na_2CO_3: strong base + weak acid: basic (1); CH_3COONa: basic (1); pH values accordingly 7, below 7, above 7 (1).	1+1+1+1+1
20	Heat baking soda to get Na_2CO_3 (1); Recrystallise to get $Na_2CO_3 \cdot 10\,H_2O$ (1); It is a basic salt (1); Used in glass, soap and paper (1); Used for removing permanent hardness (1).	1+1+1+1+1
	OR: $Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ (1); Bleaching in textile and paper (1); Disinfection of drinking water (1); Oxidising agent in chemical industry (1); Releases chlorine which kills germs (1).	1+1+1+1+1

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